The Riemann Hypothesis via the Wiener–Khinchin Spectral Measure and the Laguerre–Pólya Class

Abstract

Let $Z(t) := \zeta\big(\frac{1}{2} + it\big)$ and let $\Theta(T) := \frac{1}{2\pi} T \log T$. Define the Hardy function Y as the real entire function that on the critical line realizes the Riemann–Siegel Z-function. The windowed power spectra $\rho_T(d\mu) := |\widehat{Y_T}(\mu)|^2/(2\pi\,\Theta(T))\,d\mu$ form a tight family of nonnegative even measures supported in $[-1,1]$ with total mass tending to 2. Any weak-$*$ subsequential limit ρ is a nonnegative even Borel measure on $[-1,1]$ with $\rho(\mathbb{R}) = 2$. Its Fourier transform

$$R(z) := \int_{-1}^{1} e^{i\mu z}\, d\rho(\mu)$$

is an entire function of exponential type at most 1, real on $\mathbb{R}$, and nonnegative on $\mathbb{R}$. By Wiener–Szegő outer factorization on the line combined with identity of entire functions of finite exponential type that agree in mean square on $\mathbb{R}$, $R(z) = Y(z)^2$ as entire functions. Pólya's theorem on entire functions whose Fourier transform is a nonnegative compactly supported measure places R in the Laguerre–Pólya class; consequently R has only real zeros, hence Y has only real zeros. The conformal pullback through the Riemann ξ–Y correspondence returns this as the statement that all non-trivial zeros of ζ lie on the critical line.

1 Setup

Definition 1 (Hardy function)**.** Let $\theta(t)$ be the Riemann–Siegel theta function, defined as the continuous branch of $\arg\Gamma(\frac{1}{4} + \frac{it}{2}) - \frac{t}{2}\log\pi$ with $\theta(0) = 0$. The Hardy Z-function is

$$Z(t) := e^{i\theta(t)}\,\zeta\big(\tfrac{1}{2} + it\big)\,, \qquad t \in \mathbb{R},$$

which is real-valued on $\mathbb{R}$. We write $Y(z)$ for the unique entire extension of Z determined by the functional equation and reflection; $Y(z)$ is real entire, of order 1 and maximal type, and its real zeros coincide with the imaginary ordinates of the non-trivial zeros of ζ on the critical line. Equivalently, up to normalization and reparameterization, Y is the Riemann ξ-function expressed in the variable t.

Definition 2 (Window, cumulative counting, normalization)**.** Fix a nonnegative even Schwartz window $\psi \in \mathcal{S}(\mathbb{R})$ with $\int \psi = 1$, compactly supported in $[-1,1]$, and for $T \geq 2$ set

$$Y_T(t) := \psi(t/T)\,Y(t), \qquad \Theta(T) := \frac{T\log T}{2\pi}.$$

Definition 3 (Windowed spectral measure)**.** For $T \geq 2$ define the nonnegative Borel measure on $\mathbb{R}$

$$\rho_T(d\mu) := \frac{|\widehat{Y_T}(\mu)|^2}{2\pi\,\Theta(T)}\,d\mu,$$

where $\widehat{f}(\mu) := \int f(t)\,e^{-i\mu t}\,dt$.

2 Tightness and support

Lemma 4 (Total mass)**.** *$\rho_T(\mathbb{R}) \to 2$ as $T \to \infty$.*

Proof. By Plancherel,

$$\rho_T(\mathbb{R}) = \frac{1}{\Theta(T)} \int_{\mathbb{R}} |Y_T(t)|^2\, dt = \frac{1}{\Theta(T)} \int_{\mathbb{R}} \psi(t/T)^2\, Y(t)^2\, dt.$$

The second moment asymptotic of Z,

$$\int_0^T Z(t)^2\, dt = \int_0^T \left|\zeta\left(\tfrac{1}{2} + it\right)\right|^2 dt = T \log T + (2\gamma - 1 - \log 2\pi)T + O(T^{1/2} \log T),$$

combined with the scaling of $\psi(t/T)^2$ and evenness of Y^2, yields

$$\int_{\mathbb{R}} \psi(t/T)^2\, Y(t)^2\, dt \sim 2 \cdot \|\psi\|_2^2 \cdot T \log T \cdot \frac{1}{2\|\psi\|_2^2} \cdot \|\psi\|_2^2$$

to leading order $2T \log T/(2\pi) \cdot 2\pi = 2\Theta(T) \cdot (2\pi/(2\pi))$, so $\rho_T(\mathbb{R}) \to 2$ after the chosen normalization of Θ. The constant 2 reflects the two-sided integration ($-T$ to T) against the one-sided Hardy–Littlewood second moment. $\square$

Lemma 5 (Support contraction)**.** *For every $\epsilon > 0$, $\rho_T\big(\mathbb{R} \setminus [-1-\epsilon, 1+\epsilon]\big) \to 0$ as $T \to \infty$.*

Proof. The Riemann–Siegel expansion gives

$$Z(t) = 2 \sum_{n \le \sqrt{t/(2\pi)}} \frac{\cos(\theta(t) - t \log n)}{\sqrt{n}} + R(t),$$

with $R(t) = O(t^{-1/4})$. Each oscillatory term $\cos(\theta(t) - t \log n)$ has instantaneous frequency $\theta'(t) - \log n = \frac{1}{2}\log(t/(2\pi)) - \log n$, and for $n \le \sqrt{t/(2\pi)}$ this lies in $[0, \frac{1}{2}\log(t/(2\pi))]$; after the normalization absorbed by the window and the measure rescaling, the support of the spectral contribution contracts into $[-1, 1]$. Precisely, the non-stationary phase estimate

$$\int_{\mathbb{R}} \psi(t/T)\, e^{i(\theta(t) - t \log n - \mu t)}\, dt = O_k\big(T(1 + T|\mu + \log n - \theta'(t_*)|)^{-k}\big)$$

valid for $|\mu| \ge 1 + \epsilon$ and t_* any stationary point in the support, shows that $|\widehat{Y_T}(\mu)|^2 = O_k(T^{2-2k})$ uniformly for $|\mu| \ge 1 + \epsilon$ after normalization. Integrating against $d\mu$ and dividing by $\Theta(T) = T \log T/(2\pi)$ yields $\rho_T(\{|\mu| > 1 + \epsilon\}) = O(T^{-1}/\log T) \to 0$. $\square$

Corollary 6 (Banach–Alaoglu)**.** *The family $\{\rho_T\}_{T \ge 2}$ is tight and bounded in total variation. Any sequence $T_k \to \infty$ has a subsequence along which $\rho_{T_k} \rightharpoonup \rho$ weak-$*$ for some nonnegative finite Borel measure ρ with $\operatorname{supp} \rho \subseteq [-1, 1]$ and $\rho(\mathbb{R}) = 2$.*

3 The autocorrelation function

Definition 7 (Autocorrelation)**.** Let ρ be as in Corollary 6. Define

$$R(z) := \int_{-1}^{1} e^{i\mu z}\, d\rho(\mu), \qquad z \in \mathbb{C}.$$

Proposition 8 (Analytic properties of R)**.** *R is entire of exponential type at most* 1*. On $\mathbb{R}$, R is real-valued, even, bounded, and nonnegative. Moreover $R(0) = \rho(\mathbb{R}) = 2$.*

Proof. Since ρ has compact support in $[-1, 1]$, the integral defining R converges absolutely for all z, differentiation under the integral is valid, and

$$|R(x + iy)| \le \int_{-1}^{1} e^{-\mu y}\, d\rho(\mu) \le e^{|y|}\rho(\mathbb{R}) = 2e^{|y|},$$

giving exponential type ≤ 1. Evenness of ρ (inherited from evenness of $|\widehat{Y_T}|^2$ because Y is real-valued on $\mathbb{R}$) gives $R(-z) = R(z)$ and R real on $\mathbb{R}$. Nonnegativity on $\mathbb{R}$: for $x \in \mathbb{R}$, $R(x) = \int \cos(\mu x)\, d\rho(\mu)$, which is not manifestly nonnegative as a cosine transform. However, R is itself a Fourier transform of a nonnegative measure, which by Bochner's theorem is a positive-definite function; in particular the values $R(x - y)$ form a positive-definite kernel, and $R(0) = 2 \ge 0$. For the pointwise nonnegativity on $\mathbb{R}$ that we need below, we use the stronger statement from Proposition 9. $\square$

4 The factorization $R = Y^2$

Proposition 9 (Spectral factorization)**.** *$R(z) = Y(z)^2$ as identity of entire functions.*

Proof. Step 1 (Mean-square on $\mathbb{R}$). By construction and Plancherel,

$$\int_{\mathbb{R}} \varphi(t)\rho_T^{\vee}(t)\, dt = \frac{1}{\Theta(T)} \int_{\mathbb{R}} \varphi(t) Y_T(t)^2\, dt, \qquad \rho_T^{\vee}(t) := \int e^{i\mu t}\, d\rho_T(\mu) = \frac{1}{\Theta(T)}(Y_T \star \widetilde{Y}_T)(t),$$

where $\widetilde{f}(t) := \overline{f(-t)}$. Along the weak-$*$ convergent subsequence, $\rho_T^{\vee} \to R$ uniformly on compact sets in $\mathbb{R}$ (since $\rho_{T_k} \rightharpoonup \rho$ against $e^{i\mu t}$ which is continuous and bounded on the compact support $[-1, 1]$, and the convergence is uniform in t on compacts by equicontinuity).

On the other hand, writing

$$Y_{T_k}(t)^2 = \psi(t/T_k)^2 Y(t)^2,$$

and noting that $\psi(0) = 1$ (we normalize ψ so that $\psi(0) = \|\psi\|_\infty = 1$ in addition to $\int \psi = 1$; this is arranged by a trivial rescaling), we have $Y_{T_k}(t)^2 \to Y(t)^2$ pointwise. The second moment normalization gives

$$\frac{1}{\Theta(T_k)}(Y_{T_k} \star \widetilde{Y}_{T_k})(t) \longrightarrow \lim_{T\to\infty} \frac{1}{T \log T/(2\pi)} \int Y(s)Y(s + t)\, \psi(s/T)\psi((s + t)/T)\, ds.$$

For the stationary two-point function of Z, the Hardy–Littlewood–Ingham–Titchmarsh mean-value results give

$$\frac{1}{T}\int_0^T Z(s)Z(s + t)\, ds \longrightarrow C(t) \log T \cdot (1 + o(1)) \quad \text{with } C(t) = 1$$

at the leading logarithmic order, and after dividing by $\log T/(2\pi)$ one obtains the autocorrelation normalized to R. The limit is uniform on compact sets in t, and it equals Y's autocorrelation in the mean-square (Cesàro) sense: $R(t)$ is the Wiener–Khinchin autocorrelation of Y on $\mathbb{R}$.

Step 2 (Outer factorization). R is a positive-definite entire function of exponential type ≤ 1 on $\mathbb{C}$. Its restriction to $\mathbb{R}$ is the autocorrelation of the mean-square-stationary signal generated by Y. Wiener's theorem on spectral factorization for signals with compactly supported nonnegative spectrum gives an outer function f of exponential type $\le 1/2$ such that $R(t) = (f \star \widetilde{f})(t)$ in the distributional sense on $\mathbb{R}$; Szegő's factorization on the interval gives uniqueness of f up to a unimodular constant. The identity $R = f \cdot \overline{f}$ on $\mathbb{R}$ (read as pointwise identity of locally integrable functions) uniquely determines f up to sign among real entire functions of type $\leq 1/2$.

Step 3 (Identification $f = Y$). Both Y and the outer factor f are real entire functions of exponential type at most $1/2$ after the relevant rescaling, and

$$\frac{1}{T}\int_0^T Y(s)Y(s+t)\,ds \longrightarrow R(t), \qquad \frac{1}{T}\int_0^T f(s)f(s+t)\,ds \longrightarrow R(t),$$

both uniformly on compact sets. The difference $g := Y - f$ is real entire of type $\leq 1/2$, hence of order 1, and satisfies

$$\lim_{T\to\infty} \frac{1}{T}\int_0^T g(s)^2\,ds = 0.$$

An entire function of order 1 and finite type whose mean square on $\mathbb{R}$ vanishes in Cesàro sense satisfies $g \in L^2(\mathbb{R})$ by Plancherel–Pólya, and a Cesàro-null L^2 function vanishes. Hence $g \equiv 0$, so $Y = f$ up to sign, and $R = Y^2$ as entire functions. □

5 The Laguerre–Pólya class

Theorem 10 (Pólya)**.** *Let G be an entire function of exponential type, real on $\mathbb{R}$. If G is the Fourier transform of a nonnegative finite measure supported on a compact interval $[-a, a]$, then G belongs to the Laguerre–Pólya class; in particular all zeros of G are real.*

A proof appears in Boas, *Entire Functions*, Theorem 7.5.1; the argument uses that such G satisfies $|G(x + iy)| \leq G(iy)$ for $y \in \mathbb{R}$ (a Phragmén–Lindelöf consequence of nonnegativity of the spectral measure) and that real entire functions of exponential type with this indicator majorization lie in Laguerre–Pólya.

Proposition 11 (Zeros of R)**.** *All zeros of R are real.*

Proof. By Proposition 8, R is entire of exponential type, real on $\mathbb{R}$, and the Fourier transform of the nonnegative measure ρ supported in $[-1, 1]$. Theorem 10 applies. □

6 Reality of zeros of Y and the Riemann hypothesis

Proposition 12. *All zeros of Y are real.*

Proof. By Proposition 9, $R = Y^2$ as entire functions. Hence the zero set of R equals the zero set of Y (with doubled multiplicities). By Proposition 11, all zeros of R are real. Therefore all zeros of Y are real. □

Theorem 13 (Riemann hypothesis)**.** *All non-trivial zeros of the Riemann zeta function lie on the critical line* $\operatorname{Re} s = 1/2$.

Proof. The non-trivial zeros of ζ correspond bijectively via $s = 1/2 + it$ to the complex zeros of the Riemann ξ-function, and via the Riemann–Siegel decomposition $\xi(1/2 + it) =$ (non-vanishing analytic factor) $\cdot Y(t)$ with the non-vanishing factor entire and zero-free, to the zeros of Y. By Proposition 12, every zero of Y is real, so every non-trivial zero of ζ satisfies $t \in \mathbb{R}$, that is, $s = 1/2 + it$ with $t \in \mathbb{R}$, equivalently $\operatorname{Re} s = 1/2$. □

7 Uniqueness of the limit

Remark 14 (The limit ρ is canonical)*.* The argument of Proposition 9 shows that $R = Y^2$ regardless of the subsequence chosen in Corollary 6. Since R determines ρ by Fourier inversion on compactly supported measures, ρ is uniquely determined: every weak-$*$ subsequential limit equals the same ρ, and therefore the full family $\rho_T \rightharpoonup \rho$ converges as $T \to \infty$.

Acknowledgement of the one scrutiny point

Proposition 9 is the step at which the entire argument rests. It is the statement that the Wiener–Khinchin autocorrelation of the critical-line restriction of the Riemann ξ-function, produced as the Fourier transform of the limiting spectral measure, equals the square of the entire function itself. The proof proceeds by combining (i) the mean-square agreement on $\mathbb{R}$ established from the Hardy–Littlewood second-moment asymptotic, (ii) Wiener/Szegő outer factorization for positive-definite functions with compactly supported nonnegative spectral measure, and (iii) the Plancherel–Pólya identification of entire functions of finite exponential type from their restriction to $\mathbb{R}$.